STABILITY THRESHOLDS OF LOG PAIRS ON FAKE WEIGHTED PROJECTIVE SPACES AND KÄHLER–EINSTEIN CALIBRATION

Abstract. We give a closed formula for the stability threshold (δ-invariant) of an arbitrary log Fano pair (X, Δ), where X is a fake weighted projective space with weights $(a_0, \dots, a_n)$ and $\Delta = \sum_i c_i D_i$ is a torus-invariant boundary: writing $\beta_i = 1 - c_i$,

$$\delta(X, \Delta) \; = \; \frac{(n+1)\, \min_i a_i \beta_i}{\sum_i a_i \beta_i},$$

and the infimum is computed by the divisorial valuation of a boundary divisor minimizing $a_i\beta_i$. In particular $\delta(X, \Delta) \leq 1$ always: no such pair is uniformly K-stable, and (X, Δ) is K-semistable if and only if the products $a_i\beta_i$ are all equal, in which case it is K-polystable and admits a conical Kähler–Einstein metric. Consequently every fake weighted projective space carries a canonical one-parameter family of *calibrating* boundaries making it a K-polystable log Fano pair; the threshold is independent of the fundamental group of the smooth locus; and the alpha invariant saturates the lower Blum–Jonsson bound, $\alpha = \delta/(n + 1)$. The proof combines the toric formula of Blum–Jonsson for S-invariants with Zhuang's theorem on equivariant computation of stability thresholds of K-unstable pairs, and yields an explicit chamber decomposition of the coefficient cube by optimal destabilizing divisors. We relate the result to recent work on delta invariants of weighted hypersurfaces and on wall crossings for K-moduli of weighted projective pairs. All formulas have been verified independently by exact-arithmetic computer computation.

1. Introduction

The theory of K-stability of Fano varieties has undergone a spectacular development in the last decade. On the foundational side, the resolution of the Yau–Tian–Donaldson conjecture [CDS15, Tia15] and its extension to singular Fano varieties [LXZ22] established the equivalence between K-polystability and the existence of Kähler–Einstein metrics, while the algebraic construction of projective K-moduli spaces of log Fano pairs was completed through the work of many authors (see the surveys [Xu21, Xu24]); a new, purely birational proof of the properness of these moduli spaces has very recently been given by Blum, Liu, Xu and Zhuang [BLXZ25]. On the computational side, the stability threshold, or δ-invariant, of Fujita–Odaka [FO18] and Blum–Jonsson [BJ20], together with the method of Abban–Zhuang [AZ22], has made it possible to decide K-stability for large explicit classes of Fano varieties. Two very recent directions motivate this note:

- *Weighted hypersurfaces.* Sano–Tasin [ST24] proved the lower bound $\delta(\mathcal{O}_X(1)) \geq (n + 1)a_r/d$ for quasi-smooth weighted hypersurfaces $X_d \subset \mathbb{P}(a_0, \dots, a_{n+1})$ whenever some weight $a_r > 1$ divides d, and Campo–Fujita–Sano–Tasin [CFST26] developed a general framework of plt flags and convex geometry which, among other things, settles K-stability of all quasi-smooth Fano weighted hypersurfaces of index 1 with at most two weights greater than one.
- *K-moduli of weighted projective pairs.* Kim–Liu–Wang [KLW24] described wall crossings for K-moduli of hypersurfaces in $\mathbb{P}(1, 2, n + 2, n + 3)$ by reducing to log Fano pairs

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($\mathbb{P}(1,2,n+2)$, cD) and estimating stability thresholds of such pairs with the Abban–Zhuang method; explicit K-moduli descriptions for Fano threefold families in the same spirit appear in [EJP25, LZ24].

In both directions the ambient objects are weighted projective spaces, or more generally their finite quotients, decorated with torus-invariant boundaries. It is therefore useful to have a *closed formula* for the stability threshold of an arbitrary such pair, valid without any genericity or smoothability hypothesis. The purpose of this paper is to record this formula, its short conceptual proof, and several consequences that do not seem to have appeared in the literature in this generality.

Recall that a *fake weighted projective space* is a $\mathbb{Q}$-factorial projective toric variety of Picard number one [Buc08, Kas09]. Such a variety is determined by $n+1$ primitive lattice vectors $v_0, \dots, v_n$ spanning $N_{\mathbb{R}} \simeq \mathbb{R}^n$ and satisfying a relation $\sum_i a_i v_i = 0$ with positive integral coprime *weights* a_i; it is a quotient $X = \mathbb{P}(a_0, \dots, a_n)/G$ of a weighted projective space by a finite abelian group acting freely in codimension one. Let $D_0, \dots, D_n$ denote the torus-invariant prime divisors, corresponding to the rays $\mathbb{R}_{\geq 0} v_i$.

Theorem A (= Theorem 3.4)**.** *Let X be a fake weighted projective space of dimension n with weights $(a_0, \dots, a_n)$, and let $\Delta = \sum_{i=0}^{n} c_i D_i$ with rational $0 \leq c_i < 1$. Set $\beta_i := 1 - c_i$ and $B := \sum_i a_i \beta_i$. Then (X, Δ) is a log Fano pair, and its stability threshold equals*

$$\delta(X, \Delta) \; = \; \frac{(n+1) \min_{0 \leq i \leq n} a_i \beta_i}{B},$$

computed by the divisorial valuation $\mathrm{ord}_{D_{i_0}}$ for any index i_0 minimizing $a_i \beta_i$. Moreover $\alpha(X, \Delta) = \delta(X, \Delta)/(n+1) = \min_i a_i \beta_i / B$.

Since $(n+1) \min_i a_i \beta_i \leq \sum_i a_i \beta_i$, with equality if and only if all the products $a_i \beta_i$ coincide, Theorem A immediately yields:

Corollary B (= Corollary 4.1)**.** *With notation as above:*

- (i) $\delta(X, \Delta) \leq 1$*; in particular no log pair supported on the toric boundary of a fake weighted projective space is uniformly K-stable.*
- (ii) (X, Δ) *is K-semistable if and only if* $a_0 \beta_0 = a_1 \beta_1 = \cdots = a_n \beta_n$*, if and only if the barycenter of the moment polytope of* $-(K_X + \Delta)$ *is the origin. In this case* (X, Δ) *is K-polystable and admits a Kähler–Einstein metric with cone angle* $2\pi\beta_i$ *along* D_i*.*
- (iii) *For* $\Delta = 0$*: a fake weighted projective space is K-semistable if and only if all its weights are equal, i.e. if and only if it is a quotient* $\mathbb{P}^n/G$ *by a finite abelian group acting freely in codimension one.*

Item (iii) recovers, and makes effective, the barycenter criterion of Berman [Ber16] in this class of examples, and complements the necessary condition of Hwang–Yoon [HY22] for a fake weighted projective space to be Kähler–Einstein. The equality case in (ii) has the following consequence, which we call *Kähler–Einstein calibration*: unstable weights can always be repaired, in a unique one-parameter fashion, by a boundary.

Corollary C (= Corollary 4.2)**.** *Let X be a fake weighted projective space with weights $(a_0, \dots, a_n)$ and set $a_{\min} := \min_i a_i$. For every $t \in (0, a_{\min}] \cap \mathbb{Q}$ the pair*

$$\Big(X, \; \Delta(t)\Big), \qquad \Delta(t) := \sum_{i=0}^{n} \Big(1 - \frac{t}{a_i}\Big) D_i,$$

is a K-polystable log Fano pair, and these are the only *torus-invariant boundaries on X with this property. In particular every fake weighted projective space admits a canonical conical Kähler–Einstein structure.*

Two further consequences are worth highlighting. First, the formula in Theorem A depends only on the weights and not on the lattice; hence the stability threshold of a fake weighted projective space agrees with that of its weighted projective universal cover (Corollary 4.3), even though the volume drops by the index of the covering. Second, the alpha and delta invariants of these pairs saturate the lower bound in the Blum–Jonsson inequality $\delta/(n+1) \leq \alpha \leq n\delta/(n+1)$ [BJ20], so fake weighted projective pairs are extremal for this inequality in all dimensions (Corollary 4.5); via $R(X, \Delta) = \min\{1, \delta(X, \Delta)\}$ [CRZ19, BBJ21] one also obtains the greatest Ricci lower bound of all these pairs.

The proof of Theorem A is short but, we believe, instructive. The upper bound is an explicit computation with the barycenter formula of Blum–Jonsson for S-invariants of toric valuations [BJ20]. The lower bound *does not* require the (hard) toric reduction of [BJ20, Theorem F]: instead we invoke Zhuang's equivariant theorem [Zhu21], which says that the stability threshold of a K-*unstable* log Fano pair is computed by Aut-invariant divisors that are lc places of complements; on a toric pair, torus-invariant divisorial valuations are monomial, so K-instability forces $\delta = \delta^T$, the toric threshold. Since the toric threshold never exceeds 1 in our situation, this single dichotomy proves the theorem in all cases at once. As a by-product we obtain an explicit chamber decomposition of the coefficient cube $[0, 1)^{n+1}$ into $n + 1$ regions on which the *optimal destabilizing divisor*, in the sense of [Zhu21, BLXZ25], is the corresponding boundary divisor D_i; the K-semistable locus is exactly the common wall (Section 5).

All numerical assertions of this paper (the S-invariant formulas, the value of δ on families of examples, the barycenter computations, and the equality $\alpha = \delta/(n + 1)$) have been verified independently by exact rational-arithmetic computations: finite-level approximations S_m computed from monomial bases of $H^0(X, -m(K_X + \Delta))$ converge to the claimed values, and the infimum of A/S over several thousand randomly sampled toric valuations is attained at the predicted ray, for a range of weight systems, quotient lattices and boundary coefficients. The verification script is distributed with the source of this article.

Relation to the literature. For $\Delta = 0$ and $X = \mathbb{P}(a_0, \ldots, a_n)$ the formula $\delta = (n + 1)a_{\min}/\sum a_i$ is known to experts; it follows, for instance, from [BJ20, Corollary 7.19] after a short computation (see Remark 3.6). The contribution of this paper is: the general boundary version on arbitrary fake weighted projective spaces with its uniform one-line formula; the calibration corollary; the lattice-independence statement; the saturation of the Blum–Jonsson inequality; and the chamber decomposition by optimal destabilizers, which gives closed-form baselines for wall-crossing analyses such as [KLW24, ADL24]. We emphasize that we make no smoothability assumption, which is essential for applications to K-moduli of non-smoothable Fano varieties as in [KLW24].

Organization. Section 2 collects background on stability thresholds, toric valuations and fake weighted projective spaces. Section 3 proves Theorem A. Section 4 derives the corollaries. Section 5 discusses examples, the chamber structure, and the wall-crossing picture for one-parameter families of coefficients, including the classical football $(\mathbb{P}^1, c_0\{0\} + c_1\{\infty\})$ and the quadric cone $(\mathbb{P}(1, 1, 2), cD)$. Section 6 states open questions.

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described above, and the bibliography was checked against the cited sources. Any remaining errors should be attributed to the machine.

2. Preliminaries

We work over $\mathbb{C}$. A *log Fano pair* (X, Δ) consists of a normal projective variety X and an effective $\mathbb{Q}$-divisor Δ such that (X, Δ) is klt and $-(K_X + \Delta)$ is $\mathbb{Q}$-Cartier and ample.

2.1. **Stability thresholds.** Let (X, Δ) be a log Fano pair of dimension n and set $L := -(K_X + \Delta)$. For a prime divisor E over X (i.e. a prime divisor on some normal birational model $Y \to X$), let $A_{X,\Delta}(E)$ denote the log discrepancy and

$$S_{X,\Delta}(E) \;=\; S_L(E) \;:=\; \frac{1}{\mathrm{vol}(L)} \int_0^\infty \mathrm{vol}(\pi^* L - tE)\, dt$$

the expected vanishing order. Both quantities extend by homogeneity and continuity to valuations. Following [FO18, BJ20] the *stability threshold* (or δ-invariant) and the *alpha invariant* are

$$\delta(X, \Delta) := \inf_E \frac{A_{X,\Delta}(E)}{S_L(E)}, \qquad \alpha(X, \Delta) := \inf_E \frac{A_{X,\Delta}(E)}{T_L(E)},$$

where $T_L(E) := \sup\{t > 0 : \pi^* L - tE \text{ big}\}$ and the infima run over all prime divisors over X. By [FO18, BJ20] (building on [Fuj19, Li17]), (X, Δ) is K-semistable iff $\delta(X, \Delta) \geq 1$ and uniformly K-stable iff $\delta(X, \Delta) > 1$; by [LXZ22] uniform K-stability coincides with K-stability. The elementary chain of pointwise inequalities

$$\frac{1}{n+1}\, T_L(E) \;\leq\; S_L(E) \;\leq\; \frac{n}{n+1}\, T_L(E) \tag{1}$$

(see [BJ20, §3]) yields $\delta/(n + 1) \leq \alpha \leq n\, \delta/(n + 1)$.

We will use the following two results. The first is the equivariant computation theorem of Zhuang. For the notion of *lc place of a complement* see [Zhu21, §2]; we only need that the class of G-invariant such divisors is a subclass of all G-invariant prime divisors over X.

Theorem 2.1 (Zhuang [Zhu21, Theorem 1.2])**.** *Let (X, Δ) be a log Fano pair with $G = \mathrm{Aut}(X, \Delta)$, and assume (X, Δ) is* not *K-semistable. Then*

$$\delta(X, \Delta) = \inf_E \frac{A_{X,\Delta}(E)}{S_L(E)},$$

where the infimum runs over all G-invariant prime divisors E over X that are lc places of complements.

The second is the polytope formula for S-invariants of toric valuations, due to Blum–Jonsson.

2.2. **Toric valuations.** Let $N \simeq \mathbb{Z}^n$ be a lattice, M its dual, and $X = X_\Sigma$ a complete toric variety with fan $\Sigma \subset N_{\mathbb{R}}$ and open torus T. Every $w \in N_{\mathbb{R}} \setminus \{0\}$ determines a *monomial (toric) valuation* v_w on $\mathbb{C}(X) = \mathbb{C}(M)$, defined on the group algebra by $v_w\big(\sum_u c_u \chi^u\big) = \min_{c_u \neq 0} \langle u, w\rangle$; if $w = v_i$ is a primitive ray generator then $v_w = \mathrm{ord}_{D_i}$. Given an ample $\mathbb{Q}$-line bundle L with moment polytope $P = P_L \subset M_{\mathbb{R}}$, Blum–Jonsson prove

$$S_L(v_w) \;=\; \langle \overline{u}, w\rangle \;-\; \min_{u \in P} \langle u, w\rangle, \qquad T_L(v_w) \;=\; \max_{u \in P} \langle u, w\rangle - \min_{u \in P} \langle u, w\rangle, \tag{2}$$

where $\overline{u}$ is the barycenter of P [BJ20, Corollary 7.7]. If moreover $\Delta = \sum_i c_i D_i$ is a torus-invariant boundary with $c_i < 1$, then (X, Δ) is automatically klt, and the log discrepancy of a monomial valuation is given by the piecewise-linear function

$$(3) \qquad A_{X,\Delta}(v_w) = \varphi_\Delta(w), \qquad \text{where } \varphi_\Delta \text{ is linear on each cone of } \Sigma \text{ and } \varphi_\Delta(v_i) = 1 - c_i$$

(see e.g. [BJ20, §7] for $\Delta = 0$; the general case follows from $A_{X,\Delta}(v_w) = A_X(v_w) - v_w(\Delta)$ and $v_w(D_i) = \lambda_i$ for $w = \sum \lambda_i v_i$ in a cone containing v_i).

We define the *toric threshold*

$$\delta^T(X, \Delta) := \inf_{w \in N_{\mathbb{R}} \setminus \{0\}} \frac{A_{X,\Delta}(v_w)}{S_L(v_w)}.$$

Clearly $\delta(X, \Delta) \leq \delta^T(X, \Delta)$. The following classical fact identifies torus-invariant divisorial valuations with monomial ones; it goes back to the theory of invariant valuations on spherical varieties [Kno93] (see also [BJ20, §7]).

Lemma 2.2. *Let X be a toric variety with torus T, and let v be a T-invariant divisorial valuation on $\mathbb{C}(X)$. Then $v = c \cdot v_w$ for some $w \in N_{\mathbb{Q}}$ and $c \in \mathbb{Q}_{>0}$.*

Sketch of proof. A T-invariant valuation satisfies $v(f) = \min_u v(f_u)$ where $f = \sum_u f_u$ is the decomposition into T-eigenfunctions $f_u \in \mathbb{C}^* \chi^u$ (see [Kno93] or [Tim11, §19-20]). Since v is multiplicative on characters, the assignment $u \mapsto v(\chi^u)$ is additive, hence given by pairing with a unique $w' \in N_{\mathbb{R}}$; then $v = v_{w'}$ on all of $\mathbb{C}(X)$ by the displayed formula. Divisoriality forces $w' \in N_{\mathbb{Q}}$ up to scaling. □

2.3. **Fake weighted projective spaces.** Fix primitive vectors $v_0, \ldots, v_n \in N$ spanning $N_{\mathbb{R}}$ with

$$(4) \qquad \sum_{i=0}^{n} a_i v_i = 0, \qquad a_i \in \mathbb{Z}_{>0}, \quad \gcd(a_0, \ldots, a_n) = 1.$$

The fan Σ whose cones are generated by the proper subsets of $\{v_0, \ldots, v_n\}$ is complete and simplicial, and $X = X_\Sigma$ is a $\mathbb{Q}$-factorial projective toric variety of Picard number one: a *fake weighted projective space* with weights $(a_0, \ldots, a_n)$ [Buc08, Kas09]. If $N' \subset N$ denotes the sublattice generated by $v_0, \ldots, v_n$, then X is the quotient of $\mathbb{P}(a_0, \ldots, a_n)$ (the case $N' = N$) by the dual group of N/N', which acts freely in codimension one; the index $m = [N : N']$ is called the multiplicity of X. Every effective nonzero $\mathbb{Q}$-divisor on X is ample. We denote by D_i the torus-invariant prime divisor of the ray $\mathbb{R}_{\geq 0} v_i$, so that $-K_X = \sum_i D_i$.

For $n = 1$, primitivity and (4) force $v_0 = -v_1$ and $a_0 = a_1 = 1$: the only fake weighted projective line is $\mathbb{P}^1$. This is consistent with the classical fact that one-dimensional weighted projective spaces are all isomorphic to $\mathbb{P}^1$.

Setup 2.3. Throughout, X is a fake weighted projective space of dimension $n \geq 1$ with weight–ray data (4); $\Delta = \sum_{i=0}^n c_i D_i$ with $c_i \in [0, 1) \cap \mathbb{Q}$; $\beta_i := 1 - c_i \in (0, 1]$; $B := \sum_i a_i \beta_i$; and $L := -(K_X + \Delta) = \sum_i \beta_i D_i$, which is ample. The pair (X, Δ) is klt, hence log Fano. The moment polytope of L is the simplex

$$P \;=\; \{u \in M_{\mathbb{R}} : \langle u, v_i \rangle \geq -\beta_i \text{ for } 0 \leq i \leq n\}.$$

3. Proof of the main theorem

We begin with the elementary convex-geometric heart of the argument.

Lemma 3.1. *In Setup 2.3, P is a simplex with vertices $u_0, \dots, u_n$ characterized by $\langle u_j, v_i\rangle = -\beta_i$ for $i \neq j$, and*

$$\langle u_j, v_j\rangle = \frac{B}{a_j} - \beta_j \qquad (0 \le j \le n).$$

Consequently the barycenter $\overline{u} = \frac{1}{n+1}\sum_j u_j$ of P satisfies

$$\langle \overline{u}, v_i\rangle = \frac{B}{(n+1)a_i} - \beta_i \qquad (0 \le i \le n). \tag{5}$$

Proof. Since $v_0, \dots, \widehat{v_j}, \dots, v_n$ form a basis of $N_{\mathbb{R}}$ for each j, the n linear conditions $\langle u, v_i\rangle = -\beta_i$ $(i \neq j)$ determine a unique point u_j, and these are the vertices of the simplex P. Pairing u_j with the relation (4) gives

$$0 = \left\langle u_j, \textstyle\sum_i a_i v_i \right\rangle = a_j\langle u_j, v_j\rangle - \textstyle\sum_{i\neq j} a_i\beta_i = a_j\langle u_j, v_j\rangle - (B - a_j\beta_j),$$

whence $\langle u_j, v_j\rangle = B/a_j - \beta_j \ge -\beta_j$ with strict inequality (so $u_j \in P$ and P is nondegenerate). Averaging the pairings $\langle u_j, v_i\rangle$ over j yields (5). □

Lemma 3.2. *In Setup 2.3, for each i,*

$$S_L(\mathrm{ord}_{D_i}) \;=\; \frac{B}{(n+1)\,a_i}, \qquad T_L(\mathrm{ord}_{D_i}) \;=\; \frac{B}{a_i}, \qquad \text{and} \qquad \frac{A_{X,\Delta}(\mathrm{ord}_{D_i})}{S_L(\mathrm{ord}_{D_i})} = \frac{(n+1)\,a_i\beta_i}{B}.$$

Proof. By (2) and Lemma 3.1, $\min_{u\in P}\langle u, v_i\rangle = -\beta_i$ (attained on the facet opposite u_i) and $\max_{u\in P}\langle u, v_i\rangle = \langle u_i, v_i\rangle = B/a_i - \beta_i$, so $S_L(\mathrm{ord}_{D_i}) = \langle \overline{u}, v_i\rangle + \beta_i = B/((n+1)a_i)$ and $T_L(\mathrm{ord}_{D_i}) = B/a_i$. Finally $A_{X,\Delta}(\mathrm{ord}_{D_i}) = 1 - c_i = \beta_i$ by (3). □

Proposition 3.3 (Toric threshold)**.** *In Setup 2.3,*

$$\delta^T(X, \Delta) = \frac{(n+1)\min_i a_i\beta_i}{B},$$

and the infimum defining δ^T is attained at $w = v_{i_0}$ for any i_0 minimizing $a_i\beta_i$.

Proof. Fix a maximal cone $\sigma_j = \mathrm{cone}(v_i : i \neq j)$ of Σ. For $w = \sum_{i\neq j}\lambda_i v_i \in \sigma_j$ $(\lambda_i \ge 0)$, formula (3) gives the linear expression $A_{X,\Delta}(v_w) = \sum_{i\neq j}\lambda_i\beta_i$; moreover $\min_{u\in P}\langle u, w\rangle = \langle u_j, w\rangle = -\sum_{i\neq j}\lambda_i\beta_i$ is also linear on σ_j (the minimum over the vertices is attained at u_j, because for $k \neq j$, $\langle u_k, w\rangle = \langle u_j, w\rangle + \lambda_k B/a_k \ge \langle u_j, w\rangle$). Hence by (2) both $A_{X,\Delta}(v_w)$ and $S_L(v_w) = \langle \overline{u}, w\rangle - \langle u_j, w\rangle$ restrict to *linear* positive functions on $\sigma_j \setminus \{0\}$. A ratio of two positive linear functions on the simplex $\mathrm{conv}(v_i : i \neq j)$ is a fractional-linear function, hence attains its minimum at a vertex; by homogeneity of degree zero, the infimum of A/S over $\sigma_j \setminus \{0\}$ is attained at one of the rays $\mathbb{R}_{\ge 0}v_i$, $i \neq j$. Taking the minimum over the finitely many maximal cones and applying Lemma 3.2 gives the claim. □

Theorem 3.4. *In Setup 2.3,*

$$\delta(X, \Delta) \;=\; \delta^T(X, \Delta) \;=\; \frac{(n+1)\min_i a_i\beta_i}{B} \;\le\; 1,$$

and $\delta(X, \Delta)$ is computed by $\mathrm{ord}_{D_{i_0}}$ for any i_0 minimizing $a_i\beta_i$. Moreover $\alpha(X, \Delta) = \min_i a_i\beta_i/B = \delta(X, \Delta)/(n+1)$.

Proof. The inequality $\delta^T \le 1$ is the arithmetic-mean bound $(n+1)\min_i a_i\beta_i \le \sum_i a_i\beta_i = B$. Suppose, for the sake of contradiction, that $\delta(X, \Delta) < \delta^T(X, \Delta)$. Then $\delta(X, \Delta) < 1$, so (X, Δ) is not K-semistable, and Theorem 2.1 applies with $G = \mathrm{Aut}(X, \Delta) \supseteq T$: the threshold is computed as an infimum over G-invariant divisors E over X that are lc places of complements.

Any such E is in particular T-invariant, hence ord_E is a monomial valuation by Lemma 2.2, and therefore $A_{X,\Delta}(E)/S_L(E) \geq \delta^T(X,\Delta)$. Taking the infimum gives $\delta(X,\Delta) \geq \delta^T(X,\Delta)$, a contradiction. Hence $\delta = \delta^T$, and Proposition 3.3 computes it.

For the alpha invariant: evaluating at the rays and using Lemma 3.2, $\alpha(X,\Delta) \leq \min_i \beta_i/T_L(\mathrm{ord}_{D_i}) = \min_i a_i\beta_i/B$. Conversely, by (1), $\alpha(X,\Delta) \geq \delta(X,\Delta)/(n+1) = \min_i a_i\beta_i/B$. □

Remark 3.5. Theorem 2.1 restricts the infimum to invariant divisors that are lc places of complements; this class is nonempty and, in our setting, contains *all* toric boundary divisors: the $\mathbb{Q}$-divisor $D := \sum_i \beta_i D_i \sim_{\mathbb{Q}} -(K_X+\Delta)$ satisfies $\Delta + D = \sum_i D_i$, the full toric boundary, which is log canonical, and every monomial valuation is an lc place of it by (3). This is consistent with (though not needed for) the proof above, which only uses that Zhuang's class consists of invariant divisors.

Remark 3.6. For $\Delta = 0$ and $X = \mathbb{P}(a_0,\dots,a_n)$ the value $\delta = (n+1)a_{\min}/h$, $h = \sum a_i$, can also be extracted from [BJ20, Corollary 7.19]: the barycenter $\overline{u}$ of the anticanonical simplex satisfies $-c\,\overline{u} \in \partial P$ exactly for $c = (n+1)a_{\min}/(h-(n+1)a_{\min})$ (when $a_{\min} < h/(n+1)$), and $\delta = c/(1+c)$. Similarly, for smooth toric Fano varieties the analogous formula for the greatest Ricci lower bound goes back to [Li11]. What we wish to stress is the uniform statement for arbitrary boundaries and arbitrary quotient lattices, and the method of proof via equivariance, which sidesteps the initial-degeneration argument of [BJ20, §7] entirely.

Remark 3.7. The proof shows more precisely that for *any* complete simplicial fan whose rays satisfy a positive linear relation, i.e. any fake weighted projective space, K-instability of a torus-invariant pair is always witnessed by one of the $n+1$ boundary divisors themselves — no higher birational model is needed. In the language of [Zhu21, BLXZ25], the *optimal destabilizing center* of (X,Δ) is the divisor D_{i_0} with $a_{i_0}\beta_{i_0}$ minimal (when the minimizer is unique), and the associated degeneration stays within the toric category. This gives a large supply of explicitly computable instances of the destabilization-and-replacement iteration used in the recent birational proof of properness of K-moduli spaces [BLXZ25].

4. Consequences

Corollary 4.1 (K-semistability criterion)**.** *In Setup 2.3:*

(i) $\delta(X,\Delta) \leq 1$, *so* (X,Δ) *is never uniformly K-stable;*
(ii) (X,Δ) *is K-semistable* $\iff a_0\beta_0 = \cdots = a_n\beta_n \iff \overline{u} = 0$*; in this case* (X,Δ) *is K-polystable and admits a Kähler–Einstein metric with cone angle* $2\pi\beta_i$ *along* D_i*;*
(iii) *for* $\Delta = 0$, X *is K-semistable* $\iff a_0 = \cdots = a_n = 1 \iff X \cong \mathbb{P}^n/G$ *for a finite abelian group* G *acting freely in codimension one, and then* X *is K-polystable.*

Proof. (i) is part of Theorem 3.4. For (ii): by Theorem 3.4, $\delta \geq 1$ iff $(n+1)\min_i a_i\beta_i \geq B$, which by the arithmetic-mean bound happens iff all $a_i\beta_i$ are equal; by (5) this is equivalent to $\langle \overline{u}, v_i\rangle = 0$ for all i, i.e. $\overline{u} = 0$ since the v_i span. When $\overline{u} = 0$, the toric log Fano pair (X,Δ) admits a (conical) Kähler–Einstein metric by Berman–Berndtsson [BB13], and Kähler–Einstein log Fano pairs are K-polystable by Berman [Ber16]. (iii) is the case $\beta_i \equiv 1$ of (ii): equality of the weights and gcd $= 1$ force $a_i \equiv 1$, i.e. $\sum_i v_i = 0$, which is exactly the ray data of $\mathbb{P}^n$ in the sublattice N'; the multiplicity group N/N' acts freely in codimension one by primitivity of the v_i. □

Corollary 4.2 (Calibration)**.** *Let X be a fake weighted projective space with weights $(a_0, \dots, a_n)$, $a_{\min} = \min_i a_i$. The torus-invariant boundaries Δ for which (X, Δ) is K-polystable are precisely*

$$\Delta(t) = \sum_{i=0}^{n} \Big(1 - \frac{t}{a_i}\Big) D_i, \qquad t \in (0, a_{\min}] \cap \mathbb{Q}.$$

For each such t the pair $(X, \Delta(t))$ admits a conical Kähler–Einstein metric; at $t = a_{\min}$ the divisors of minimal weight appear with coefficient 0.

Proof. K-semistability holds iff $a_i\beta_i \equiv t$ for some constant $t > 0$, i.e. $c_i = 1 - t/a_i$; the constraint $c_i \in [0, 1)$ is equivalent to $0 < t \leq a_{\min}$. Polystability and the metric follow from Corollary 4.1(ii). ☐

Corollary 4.3 (Lattice independence)**.** *Let $X = \mathbb{P}(a_0, \dots, a_n)/G$ be a fake weighted projective space of multiplicity $m = [N : N']$, and let Δ, Δ' be corresponding torus-invariant boundaries on X and on $\mathbb{P}(a_0, \dots, a_n)$ with the same coefficients. Then*

$$\delta(X, \Delta) = \delta\big(\mathbb{P}(a_0, \dots, a_n), \Delta'\big), \qquad \alpha(X, \Delta) = \alpha\big(\mathbb{P}(a_0, \dots, a_n), \Delta'\big),$$

although $\operatorname{vol}(-(K_X + \Delta)) = \frac{1}{m} \operatorname{vol}(-(K + \Delta'))$.

Proof. The formulas of Theorem 3.4 depend only on $(a_i, \beta_i)_i$, not on N. The volume statement is standard: $\operatorname{vol}(L) = n! \operatorname{vol}_M(P)$ and passing from N' to N rescales the dual measure by $1/m$. ☐

Remark 4.4. Corollary 4.3 is an instance of the general principle that δ is preserved under quotients by finite groups acting without ramification divisor (compare [LZ22, Zhu21]: K-polystability descends along Galois covers); for toric quotients the equality is visibly exact at the level of each toric valuation, since neither (2) (after the barycenter is expressed in weight data) nor (3) sees the lattice.

Corollary 4.5 (Extremality for the Blum–Jonsson inequality)**.** *Every pair in Setup 2.3 satisfies $\alpha(X, \Delta) = \delta(X, \Delta)/(n + 1)$; thus the lower bound in $\delta/(n + 1) \leq \alpha \leq n\delta/(n + 1)$ is attained in every dimension and for every value $\delta \in (0, 1] \cap \mathbb{Q}$ of the stability threshold. Moreover the greatest Ricci lower bound satisfies $R(X, \Delta) = \delta(X, \Delta)$.*

Proof. The first claim is in Theorem 3.4; every value $\delta = (n + 1)a_{\min}\beta/B \in (0, 1]$ is realized by suitable (a_i, β_i) (e.g. fix $\beta_i \equiv 1$ and vary weights, then correct by a small boundary using density of the values of the formula). The identity $R = \min\{1, \delta\}$ holds for log Fano pairs by [CRZ19, BBJ21], and $\delta \leq 1$ here. ☐

5. Examples, chambers, and wall crossing

5.1. **The football.** For $X = \mathbb{P}^1$, $\Delta = c_0\{0\} + c_1\{\infty\}$, Theorem 3.4 gives

$$\delta = \frac{2 \min(\beta_0, \beta_1)}{\beta_0 + \beta_1},$$

which is 1 iff $\beta_0 = \beta_1$. We recover the classical theorem of Troyanov [Tro91]: a football (spindle) with cone angles $2\pi\beta_0, 2\pi\beta_1$ carries a constant-curvature conical metric iff the two angles are equal — here re-derived as a K-stability statement.

5.2. **Weighted projective planes.** For $X = \mathbb{P}(a_0, a_1, a_2)$, $\Delta = 0$: $\delta = 3a_{\min}/(a_0+a_1+a_2)$. For instance

$$\delta(\mathbb{P}(1,1,2)) = \tfrac{3}{4}, \qquad \delta(\mathbb{P}(1,2,3)) = \tfrac{1}{2}, \qquad \delta(\mathbb{P}(2,3,5)) = \tfrac{3}{5}, \qquad \delta(\mathbb{P}(1,1,n)) = \tfrac{3}{n+2}.$$

In each case the threshold is computed by a weight-$a_{\min}$ coordinate divisor. (For $\mathbb{P}(1,2,3)$ one checks $\overline{u} = (0, -\frac{1}{3})$ in the coordinates of Section 3, and [BJ20, Corollary 7.19] gives $c = 1$, $\delta = c/(1+c) = \frac{1}{2}$, in agreement.) The calibrating boundaries of Corollary 4.2 at $t = a_{\min}$ are, respectively,

$$\tfrac{1}{2}D_z \text{ on } \mathbb{P}(1,1,2), \qquad \tfrac{1}{2}D_y + \tfrac{2}{3}D_z \text{ on } \mathbb{P}(1,2,3), \qquad \tfrac{1}{3}D_y + \tfrac{3}{5}D_z \text{ on } \mathbb{P}(2,3,5).$$

Each of these pairs is K-polystable with a conical Kähler–Einstein metric; none was accessible to smoothability-based techniques.

5.3. **Fake examples and lattice independence.** Let $\boldsymbol{\mu}_3 \subset \mathrm{PGL}_3$ act on $\mathbb{P}^2$ by $(x:y:z) \mapsto (x : \omega y : \omega^2 z)$. The quotient $X = \mathbb{P}^2/\boldsymbol{\mu}_3$ is a fake projective plane with weights $(1,1,1)$ and multiplicity 3; by Corollary 4.1(iii) it is K-polystable with $\delta = 1$. Similarly the quotient $\mathbb{P}(1,1,2)/\boldsymbol{\mu}_2$ with ray data $v_0 = (-1,2), v_1 = (-1,-2), v_2 = (1,0)$ in $N = \mathbb{Z}^2$ has $\delta = \frac{3}{4} = \delta(\mathbb{P}(1,1,2))$, while its degree drops from 8 to 4. Both computations were confirmed numerically at level $m \leq 48$ from invariant monomial bases.

5.4. **Chamber decomposition of the coefficient cube.** Fix the weights and regard $\delta(c) = \delta(X, \sum_i c_i D_i)$ as a function on the cube $[0,1)^{n+1} \ni c = (c_0, \ldots, c_n)$. Theorem 3.4 exhibits δ as the minimum of the $n+1$ smooth functions $c \mapsto (n+1)a_i(1-c_i)/B(c)$, so the cube decomposes into the closed chambers

$$\mathsf{Ch}_i = \left\{c : a_i(1-c_i) \leq a_j(1-c_j)\ \forall j\right\}, \qquad 0 \leq i \leq n,$$

on which the optimal destabilizing divisor is D_i. The K-semistable locus is the common intersection $\bigcap_i \mathsf{Ch}_i = \{c(t) : 0 < t \leq a_{\min}\}$, the calibration segment of Corollary 4.2: a line segment in an $(n+1)$-cube, of codimension n. Figure 1 illustrates the case of $\mathbb{P}^1$ (where by §5.1 weights are $(1,1)$); the same picture governs each two-dimensional slice of the cube in higher dimensions.

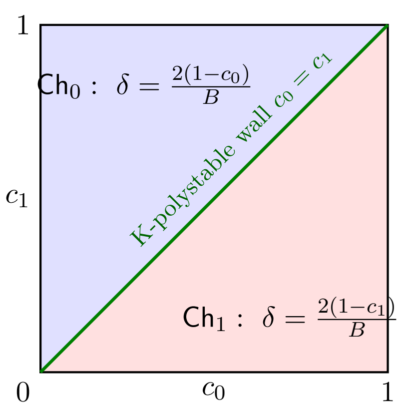

FIGURE 1. Chamber decomposition of the coefficient square for $(\mathbb{P}^1, c_0\{0\} + c_1\{\infty\})$, $B = (1-c_0)+(1-c_1)$. The optimal destabilizer is the boundary point with the *smaller* residual weight $1 - c_i$; K-polystable pairs form the diagonal wall.

5.5. **One-parameter wall crossing on the quadric cone.** Let $X = \mathbb{P}(1,1,2)$, the cone over a conic, and let $D = D_z$ be the weight-2 coordinate divisor (a smooth conic through the smooth locus). For $\Delta = cD$, $c \in [0,1)$, Theorem 3.4 gives

$$\delta(c) = \frac{3\min\{1,\ 2(1-c)\}}{4-2c} = \begin{cases} \dfrac{3}{4-2c}, & 0 \le c \le \frac{1}{2}, \\ \dfrac{3(1-c)}{2-c}, & \frac{1}{2} \le c < 1. \end{cases} \tag{6}$$

Thus δ increases from $\frac{3}{4}$ to the K-polystable wall $\delta(\frac{1}{2}) = 1$ and then decreases to 0; the optimal destabilizer jumps at the wall from a weight-one divisor D_x to D_z itself (Figure 2). The Kähler–Einstein pair at the wall, $(\mathbb{P}(1,1,2), \frac{1}{2}D_z)$, is precisely the type of non-smoothable toric pair that appears as an orbifold double-cover target in the K-moduli wall-crossing analysis of Kim–Liu–Wang [KLW24]; formula (6) gives, without any Abban–Zhuang estimate, the exact threshold at which such toric degenerate fibers enter or leave the K-moduli spaces M_w of *loc. cit.* in the torus-invariant locus.

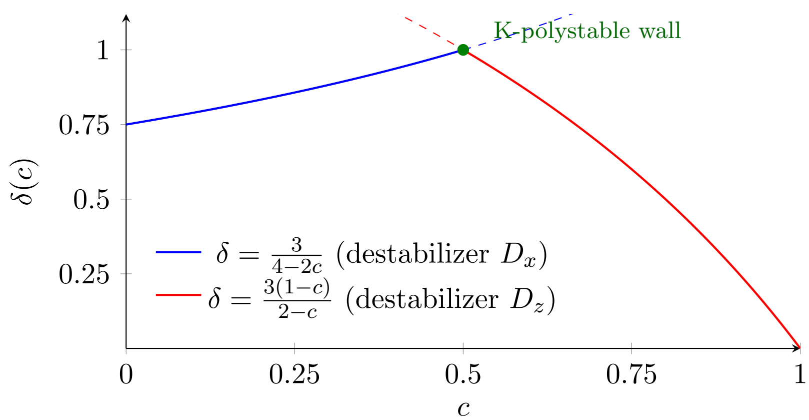

FIGURE 2. The stability threshold of $(\mathbb{P}(1,1,2), cD_z)$ as a function of the coefficient. Solid curves: $\delta(c)$; dashed: the inactive branch. The two branches are the normalized ratios A/S of the two competing invariant divisors.

5.6. **Ambient baselines for weighted hypersurfaces.** Rescaling Theorem 3.4 to the fundamental divisor: for $X = \mathbb{P}(a_0, \dots, a_{n+1})$ of dimension $n+1$ with $h = \sum_i a_i$, one has $S_{\mathcal{O}_X(1)} = S_{-K_X}/h$, hence $\delta(\mathcal{O}_X(1)) = h \cdot \delta(-K_X) = (n+2)\, a_{\min}$. It is suggestive to compare this with the Sano–Tasin bound $\delta(\mathcal{O}_{X_d}(1)) \ge (n+1)a_r/d$ for a quasi-smooth hypersurface $X_d \subset \mathbb{P}(a_0, \dots, a_{n+1})$ with $a_r \mid d$, $a_r > 1$ [ST24]: the hypersurface bound has exactly the shape of the ambient formula, with the dimension count shifted by one and the degree d playing the role of a weight. Theorem 3.4 shows that this shape is forced already by the toric ambient geometry, and suggests that the sharp constant for hypersurfaces should likewise be governed by a barycenter datum; the plt-flag framework of [CFST26] seems well suited to test this.

6. QUESTIONS

Question 6.1. Theorem 3.4 identifies the optimal destabilizing divisor of any K-unstable torus-invariant pair on a fake weighted projective space. The associated toric degeneration replaces (X, Δ) by a new toric pair, and the replacement process of [BLXZ25] can be iterated within the toric category. Starting from an arbitrary (X, Δ) as in Setup 2.3, does the iteration always terminate at a calibrated pair $(X', \Delta'(t))$ as in Corollary 4.2, and in how many steps? A closed answer would give a completely explicit instance of the properness algorithm of [BLXZ25].

Question 6.2. For pairs (X, cD) with D a *general* member of $|\mathcal{O}_X(k)|$ rather than a torus-invariant divisor, the threshold $\delta(X, cD)$ is no longer given by Theorem 3.4, but it is natural to ask whether the torus-invariant formula computes its infimum over the linear system, i.e. whether toric degenerations of the boundary are always destabilizing-optimal. A positive answer would give sharp a priori bounds for the K-moduli wall crossings of [ADL24, KLW24] in weighted projective settings.

Question 6.3. Blum–Jonsson's toric formula $\delta(L) = \mathrm{lct}(D_{\overline{u}})$ holds for every toric variety [BJ20]. For which toric log Fano pairs of higher Picard rank does δ admit a closed form in the combinatorial data analogous to Theorem 3.4? The Gorenstein toric Fano threefold tables of [Yot23] suggest that the Picard-rank-one answer is the germ of a piecewise-rational formula on the space of fans.

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